

Accuracy as a Fraction

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Training and Testing

Without opening the old lesson, explain one key idea from Training and Testing. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Features and Distractors

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Debugging Strategies

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Compare both class fractions for Accuracy and class imbalance. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Accuracy and class imbalance. Proposed explanation: “High accuracy means both classes work well”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Assess a new comparison independently

New training labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records. The always-clear baseline is $[[18,0],[2,0]]$ and candidate is $[[15,3],[1,1]]$, with reference rows and clear then blocked on both axes. Calculate ordinary accuracy, both class fractions and balanced accuracy for each. Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Class A gets 8/10 cases right and class B gets 18/30 right. Compute each rate and the combined rate. Why is averaging the two percentages misleading here?

6. Solve the given case

Nine of twelve held-out classifications are correct.

Compute accuracy and state the denominator.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A classifier gets 9 of 12 cases right. Another gets 8 of 10 right. Calculate both accuracies and identify the larger fraction.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 12 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Report correct recommendations / all test users, alongside the raw test count.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Class A gets 8/10 cases right and class B gets 18/30 right. Compute each rate and the combined rate. Why is averaging the two percentages misleading here?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Score the majority baseline

Training has 16 clear and 4 blocked labels. State the majority rule. On the worked evaluation, give its matrix, accuracy, false positives and false negatives with blocked positive. Explain why 80 percent is incomplete.

My reasoning and evidence

13. W2 Compare both class fractions

For the worked candidate $[[12,4],[0,4]]$, give ordinary accuracy, both class-specific correct fractions, balanced accuracy, false positives and false negatives. Compare with the baseline.

My reasoning and evidence

14. W3 Explain a changed class mix

For the illustrative 8-clear 8-blocked set, use baseline $[[8,0],[8,0]]$ and candidate $[[6,2],[0,8]]$. Calculate ordinary and balanced accuracy for each. Explain what stayed fixed and what this example does not prove.

My reasoning and evidence

15. W4 Assess a new comparison independently

New training labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records. The always-clear baseline is $[[18,0],[2,0]]$ and candidate is $[[15,3],[1,1]]$, with reference rows and clear then blocked on both axes. Calculate ordinary accuracy, both class fractions and balanced accuracy for each. Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

My reasoning and evidence

16. W5 Limit claims from missing classes

An evaluation contains ten clear cases and no blocked cases. All predictions are clear. State ordinary accuracy, explain why blocked performance is unavailable, and describe how our code handles the missing class.

My reasoning and evidence

17. Summaries with counts and proportions — W1 State the denominator

In the worked poll, why is the known denominator 4 rather than 5?

My reasoning and evidence

18. Summaries with counts and proportions — W2 Compare proportions

Poll A has 2 Yes of 4 known and Poll B has 3 Yes of 6 known. Compare proportions and counts.

My reasoning and evidence

19. Summaries with counts and proportions — W3 Repair zero division

All three responses are Unknown. What should the program report?

My reasoning and evidence

20. Summaries with counts and proportions — W4

Summarize independently

Process Yes, Yes, No, Unknown, No, Yes. Report Yes count, known count, unknown count and Yes proportion among known.

My reasoning and evidence

21. Summaries with counts and proportions — W5 Avoid overclaiming

Three of five known volunteers said Yes. Can you say 60 percent of all students agree?

My reasoning and evidence

22. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

23. Pictorial case practice: Accuracy as a Fraction

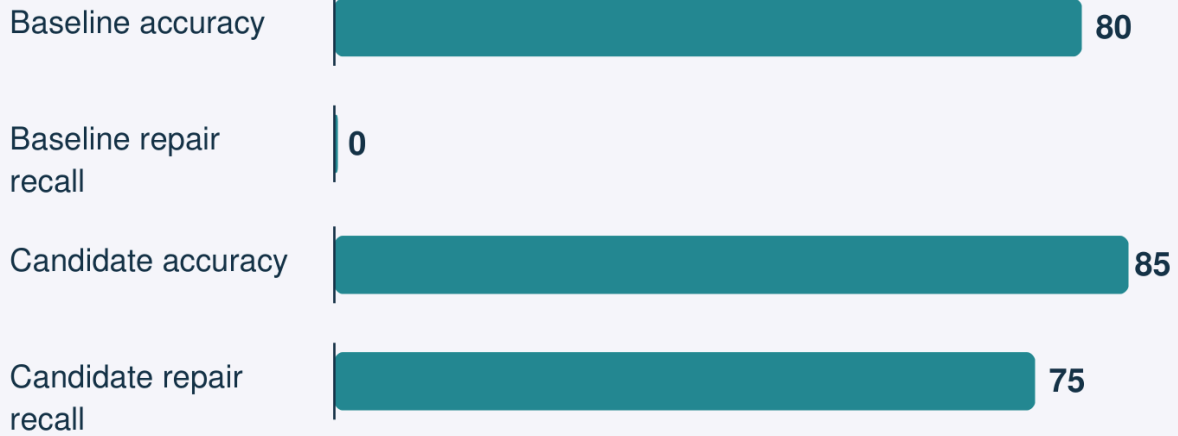
A fictional shelf classifier predicts repair needed or not needed.

Evaluation reference labels: 16 not-needed and 4 needed. Positive class is needed.

Always-not-needed baseline is correct on 16. Candidate has $TP=3, FN=1, FP=2, TN=14$.

Accuracy can hide missed repair cases

percent



Bar length shows the supplied value; the marked baseline is zero.

Accuracy as a Fraction

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Reference population

20 labelled cases, including four repair-needed books.



2. Baseline

$16/20=80\%$ accuracy; catches 0/4 needed cases.



3. Candidate

$(3+14)/20=85\%$ accuracy; catches $3/4=75\%$ needed cases.



4. Compare costs

Candidate also raises two false alarms; discuss both missed repairs and extra checks.

CHANGE ONE THING

A candidate instead marks every book repair needed.

Explain the changed result using the matching lesson.

Why is 100% repair detection not sufficient to declare the changed classifier best?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate accuracy.

24. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Compare recommender choices with a simple baseline and report relevant rare-category failures, not only aggregate accuracy.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

25. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional shelf classifier predicts repair needed or not needed.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Candidate also raises two false alarms; discuss both missed repairs and extra checks.

1. Reference population
2. Compare costs
3. Candidate
4. Baseline

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compare costs Candidate also raises two false alarms; discuss both missed repairs and extra checks. A majority-class baseline can look accurate while missing every positive case. The candidate improves overall agreement and repair detection in this table, but false alarms still matter. Use the same cases, denominators and positive-class definition for both. Do not infer that a single metric identifies the best policy for every library.

Practice 2: explain the deciding evidence

Changed-case practice: A candidate instead marks every book repair needed. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Recall is $4/4=100\%$, but accuracy is only $4/20=20\%$.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Recall is $4/4=100\%$, but accuracy is only $4/20=20\%$. All 16 not-needed books become false alarms.

Repair a missing building block

Review Training and Testing (ai-2/11). Goal: Choose a split unit that matches the intended prediction claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan counts?. Return to this week and explain what you changed.

Review Features and Distractors (ai-2/9). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Does this one pair prove the classifier always ignores subject? Explain a narrower claim.. Return to this week and explain what you changed.

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
